INCH-POUND

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MILITARY SPECIFICATION

BEEF STRIPS WITH GREEN PEPPERS AND GRAVY, THERMOSTABILIZED, TRAY PACK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers beef strips with green peppers and gravy, thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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SPECIFICATIONS

FEDERAL

PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-L-1497	-	Labeling of Metal Cans for Subsistence Items
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Personal Equipment and Equipage; General Specification for
MIL-C-44340	-	Can, Tray Pack

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-900	-	Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Thermostabilized Foods for the Armed Forces

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies are available from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Institutional Meat Purchase Specifications for Fresh Beef, Series 100

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(Copies are available from the Director, Livestock and Seed Division, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2603, South Building, P. O. Box 96456, Washington, DC 20090-6456.)

Meat and Poultry Inspection Regulations (9 CFR Parts 301-399)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

U.S. Standards for Condition of Food Containers

(Copies are available from the Chairperson, Condition of Container Committee, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2506, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS), U.S. FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, One Maritime Plaza, 23rd Floor, San Francisco, CA 94111.)

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regulations unless a specific exemption has been obtained. laws and

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3330 - Peel Adhesion of Pressure-Sensitive Tape

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AOAC INTERNATIONAL

Official Methods of Analysis of the AOAC

(Application for copies should be addressed to AOAC International, 2200 Wilson Boulevard, Suite 400, Arlington, VA 22201-3301.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Beef. Beef shall be from steers, heifers, or cows and shall be derived from one or any combination of the following USDA Certified Institutional Meat Purchase Specifications (IMPS) Cuts: Item No. 114 (shoulder clod), Item No. 161 (round, shank-off, boneless), Item No. 163 (round, shank-off, 3-way, boneless), Item No. 165 (round, rump, and shank-off, boneless), Item No. 167A (round, knuckle, peeled), Item No. 169 (top round), Item No. 170A (bottom round, heel-out), and Item No. 171B (outside round). All beef shall be certified by a USDA Agricultural Marketing Service (AMS) agent for condition of the product and compliance with the IMPS Quality Assurance Provisions.

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NOTE: Certified IMPS cuts may be further trimmed (see 3.3.1) and broken down into smaller cuts after certification, as requested by the purchaser. Boxes used for shipment of the further trimmed and broken down cuts shall be labeled as "Beef for further processing as specified in MIL-B-44228". The certificate shall be noted as such and shall accompany the shipment.

3.2.1.1 Handling and storage. Handling and storage of the boned and trimmed beef, prior to processing into the finished product, shall be in accordance with the following requirements:

a. Beef processed on the day of initial certification shall be maintained in the temperature range of 28°F to 50°F, inclusive.

b. Holding in the fresh-chilled state for not more than 4 days after initial certification is permitted, provided that the beef is maintained in the temperature range of 28°F to 40°F, inclusive.

c. Holding in the frozen state for not more than 180 days after initial certification is permitted, provided that the beef is:

- frozen to 0°F or lower within 72 hours after initial certification
- stored at 0°F or lower
- protected from freezer deterioration and damage
- stored in containers that are adequate to maintain product excellence

d. Tempering/thawing of frozen beef shall comply with the following:

- microwave tempering/thawing of frozen beef is permitted provided excellent condition of the beef, as determined by the USDA, is maintained
- the microwave tempering/thawing process shall not exceed 48 hours from time of removal of the beef from the freezer to further preparation
- regardless of the process used, frozen beef shall be tempered/thawed to a temperature not to exceed 40°F
- when a tempering/thawing process other than microwave is used, the time from the start of the tempering/thawing process (removal from freezer) to further preparation shall not exceed 96 hours
- beef which has been tempered/thawed shall not be refrozen

3.2.2 Water. Water used for formulation, ice making, and washing shall conform to the National Primary Drinking Water Regulations.

3.2.3 Starch, food, modified, high opacity. The high opacity starch shall be a white, odorless, finely pulverized, modified waxy maize food starch for use in thermostabilized foods. The modified starch shall demonstrate initial

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viscosity development in the temperature range of 140°F to 170°F and be fully hydrated at common retort temperatures. The starch shall resist breakdown at low pH, under shear stress, and under conditions of cold storage. The cooked starch shall be bland with essentially no cereal or starch taste.

3.2.4 Soup stock, dehydrated. The dehydrated soup stock shall be a light brown to tan, free flowing powder derived from liquid soup stock. The soup stock shall be made from USDA inspected beef bones, or pork bones, with adhering meat, or a combination of the two materials, and shall have a minimum of 93 percent total solids and a maximum of 2 percent fat.

3.2.5 Sauce, soy. The soy sauce shall have been prepared from water, wheat, soybeans, and salt. The soy sauce shall contain not more than 0.10 percent sodium benzoate as a preservative and shall have a titratable acidity of not less than 0.95 percent or more than 1.15 percent. The flavor shall be tart, sweet, salty, with a definite fermented bouquet. The soy sauce shall be a clear, reddish brown color with no visual impurities. The soy sauce shall be of the latest date of pack.

3.2.6 Salt. Salt shall be noniodized, white, refined sodium chloride with or without anticaking agents and shall comply with the purity standards for sodium chloride of the Food Chemicals Codex.

3.2.7 Sugar, white, granulated. Sugar shall be white, refined, sucrose, granulated cane or beet, or a combination thereof and shall comply with MIL-STD-900.

3.2.8 Lecithin. Lecithin shall comply with the Food Chemicals Codex.

3.2.9 Garlic powder. Garlic powder shall be Fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.10 Pepper, black, ground. The ground black pepper shall have been ground from the deep brown to black, deep-set, wrinkled, immature berries of Piper nigrum L. The ground pepper shall have a characteristic, penetrating odor, a hot biting pungent flavor, and a light gray to speckled black-gray color. The black pepper shall contain not less than 2.0 mL of volatile oil per 100 grams and be of such size that 95 percent shall pass through a U.S. Standard No. 16 sieve.

3.2.11 Pepper, green, sweet, dehydrated. The peppers shall be 1/4 inch dices of sweet bell peppers that have been air-dried, freeze-dried, or a combination of both. The pepper dices shall be practically free of seeds, seed cells, and diseased or rotted areas. The pepper dices shall possess a fresh, clean, typical sweet green pepper flavor and odor and a green (with minimal brownish) color. The moisture content shall be not greater than 5 percent.

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3.2.12 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the requirements of this specification. The containers used for the spice and seasoning blend shall be labeled with each ingredient and the percentage of each ingredient in the blend. The ingredients shall be in the same proportions as specified in the ingredient formula.

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Beef preparation. The beef shall be prepared as follows:

a. Boned and trimmed beef meeting the requirements of 3.2.1 shall be further trimmed, if necessary, to ensure compliance with the finished product requirements (see 3.6).

b. The boneless beef shall be mechanically cut with cutter settings of 1/2 inch by 1/2 inch by 3 inches (± 1/8 inch in each dimension). The beef strips shall be cooked by a commercially acceptable cook method to produce a product meeting finished product requirements (see 6.4). (Water cooking (blanching) of the beef strips in a minimum amount of water is permitted provided all finished product requirements are met.) Retain broth and rendered fat, if possible with cook method used, for inclusion in the gravy. The broth and rendered fat shall be incorporated in the gravy formulation immediately following cooking of the beef and used in the maximum amount available. Alternatively, whole beef cuts may be oven roasted or otherwise cooked by a commercially acceptable method and then cut into strips as specified above. The process of retaining the broth and rendered fat also applies. Whole beef cuts shall not be water cooked.

c. The cooked beef shall be maintained in the temperature range of 28°F to 40°F for not more than 48 hours from time of cooking to time of filling into tray pack cans. Cooling is not necessary if filling occurs immediately after cooking.

3.3.2 Gravy preparation. The gravy shall be formulated and prepared as follows:

Ingredients	Percent by weight
Water and broth	86.37
Starch, food, modified, high opacity 1/	6.00
Soup, stock, dehydrated	3.00
Sauce, soy	2.75
Salt 2/	0.75
Sugar, white, granulated	0.75
Lecithin	0.25
Garlic powder	0.08
Pepper, black, ground	0.05

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1/ The total amount of starch in the gravy formula may be adjusted, if necessary, to ensure compliance with the finished product viscosity requirements (see 3.6).

2/ The total amount of salt in the gravy formula shall be adjusted as necessary to produce a product that complies with the finished product salt requirement (see 3.6).

NOTE: The following gravy preparation procedures were used in the development of this product. Alternate procedures may be used provided finished product requirements are met. (When alternate procedures are used, the time and temperature requirements specified for the prepared gravy are still applicable.)

a. A slurry shall be made with the starch and part of the water.

b. The remaining ingredients shall be mixed together and heated to not less than 140°F.

c. The starch slurry shall be added and the final mixture heated to 180°F to 190°F and held for approximately five minutes at this temperature.

d. The volume of the final mixture shall be adjusted with water to compensate for evaporation loss during heating and holding.

e. The gravy shall be maintained in the temperature range of 150°F to 180°F for not more than four hours from time to preparation to time of filling into cans (see 3.4).

3.3.3 Product preparation. The beef strips and gravy shall be combined in the following approximate proportions. The percent by weight of the beef strips and percent by weight of the gravy may be adjusted, if necessary, to ensure compliance with the finished product drained weight requirements.

Ingredients	Percent by weight
Beef strips, cooked	60.38
Gravy	38.87
Peppers, green, sweet, dehydrated	0.75

3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with product to conform to the finished product requirements and to the following requirements:

a. A single or multi-stage filling operation may be used.

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b. Immediately after filling, each can shall be sealed in accordance with the can manufacturer's guidelines/requirements and 21 CFR, Part 113, Subpart D, or 9 CFR, Part 318, Subpart G, as applicable (see 4.5.5), and under a vacuum established by a processing authority and specified in the scheduled process so as to ensure compliance with the finished product requirement (see 3.6s).

c. Each filled and sealed tray pack can shall be in the retort process within 2 hours after sealing.

3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting until a sterilization value (F_o) or not less than 6 has been achieved.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, or moldy.

c. There shall be no color foreign to the product.

d. No individual can shall contain less than 104 ounces.

e. The average net weight shall be not less than 106 ounces.

f. No individual can shall contain less than 47.0 ounces drained weight of beef pieces and green peppers (combined).

g. The average drained weight of product shall be not less than 51.0 ounces.

h. Total weight of cartilage, coarse connective tissue, sections of tendons or ligaments, and glandular material, collectively, in any individual can shall be not more than 2.0 ounces.

i. No individual can shall contain a bone piece measuring 0.3 inch or more in any dimension.

j. There shall be not less than 50.0 percent (by weight) of beef strips measuring 1.0 inch or more in the longest dimension.

k. The texture of the beef strips shall not be dry, tough, rubbery, or mushy.

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l. The green peppers shall be rehydrated but shall not be mushy, hard, fibrous, or tough.

m. Gravy shall not be lumpy.

n. There shall be a uniform mixture of beef and sauce.

o. The product shall show no evidence of excessive heating (materially darkened or scorched).

p. The average fat content of the finished product shall be not greater than 7.0 percent and no individual tray pack shall have a fat content greater than 9.0 percent.

q. No individual tray pack shall have a salt content greater than 1.0 percent or less than 0.5 percent.

r. The viscosity of the gravy shall be not less than 7.0 cm per 10 seconds nor greater than 20.0 cm per 10 seconds when determined by a Bostwick Consistometer (see 4.5.3.4).

s. Filled, sealed, and retorted cans shall show evidence of proper vacuum as determined by concavity of the can lid (see 4.5.8).

3.6.1 Palatability. The finished product shall be equal to or better than the preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The beef components and the finished product shall originate and be produced, processed, and stored in plants regularly operating under the Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all the requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection and supervision necessary to ensure compliance with the requirements of this specification.

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4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material examination. In accordance with 4.1, components and materials shall be examined in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the AOAC International and in the Approved Methods of the American Association of Cereal Chemists to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Tray pack inspection. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.5.3.1 Net weight examination. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing the total weight by 30 from the weight of each tray pack in the sample.

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The results shall be reported to the nearest 1 ounce. If the average net weight is less than 106 ounces or if the net weight of any individual can is less than 104 ounces, the lot shall be rejected.

4.5.3.2 Double sampling plan for product examination. The finished product shall be examined for the defects listed in table I utilizing the double sampling plans indicated in MIL-STD-105. The lot size shall be expressed in tray pack cans. The sample unit shall be one filled and sealed tray pack can. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0 for major defects. The sample cans shall be heated in accordance with the heating instructions on the can label.

TABLE I. Product defects 1/ 2/

Category	Defect
Major	
101	Drained weight of beef pieces and green peppers in a can is less than 47.0 ounces 3/ 4/
102	Total weight of cartilage, coarse connective tissue, sections of tendons or ligaments, and glandular material, collectively, in a can is more than 2.0 ounces
103	Presence of a bone piece measuring 0.3 inch or more in any dimension
104	More than 50.0 percent (by weight) of beef strips less than 1 inch in length 3/
105	Beef strips are dry, tough, rubbery, or mushy
106	Green peppers not rehydrated or are mushy, hard, fibrous, or tough
107	Gravy is lumpy
108	Not a uniform mixture of beef pieces and gravy
109	Product shows evidence of excessive heating (materially darkened or scorched)

1/ The presence of foreign material (for example, dirt, insect parts, hair, wood, glass, metal), foreign odor or flavor (for example, burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

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2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot.

3/ To determine drained weight, the gravy in the can shall be poured off, strained through a U.S. Standard No. 8 sieve, and reserved for viscosity determinations. The remaining contents shall be poured into a flat-bottom container. A minimum of three times the tray pack can's volume of 180°F to 190°F water shall be added to the container so as to cover the contents. The contents and water shall be agitated so as to liquify rendered fat and remove the gravy without breaking the beef pieces and peppers. The contents shall then be poured into a U.S. Standard No. 8 sieve in a manner that will distribute the product over the sieve without breaking the beef pieces and peppers. The sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted at approximately a 45° angle and allowed to drain for 2 minutes before determining the drained weight by subtracting the sieve tare weight from the gross weight. The drained weight shall be reported to the nearest 0.1 ounce.

4/ The lot shall be rejected if the sample average drained weight is less than 51.0 ounces.

4.5.3.3 Fat and salt content testing. Three filled and sealed tray pack cans shall be selected at random from the lot. The product shall be tested for fat content in accordance with the Official Methods of Analysis of the AOAC, methods 976.21, 985.15, or 960.39, and for salt content in accordance with the Official Methods of Analysis of the AOAC, method 935.47, except that preparation of the samples shall be as follows: The unopened tray pack cans shall be gently warmed in a water bath to melt fat adhering to the inside of the cans. The cans shall be opened and the entire contents of each can shall be separately blended in a Waring Blendor or equivalent. The test results shall be reported to the nearest 0.1 percent. Any result failing to conform to the fat or salt requirements in 3.6 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.3.4 Viscosity testing. The strained gravy collected from each of the cans in the first sample of cans selected for drained weight inspection (see 4.5.3.2 and footnote 3/ to table I) shall be individually tested for viscosity as specified below (see 6.5).

Instrument: Bostwick Consistometer

Catalog number: 23270-004 or Catalog number: 15-347-50
VWR Scientific Company Fisher Scientific
P.O. Box 7900 585 Alpha Drive
San Francisco, CA 94120 Pittsburgh, PA 15238

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Method:

a. Level the instrument.

b. Bring gravy to 100°F ± 1°F in a water bath in a covered container.

c. Stir gravy thoroughly before filling the Bostwick cavity.

d. Scrape gravy evenly across upper edge of cavity.

e. Release gravy and time gravy flow to the nearest one second and measure distance traveled to the nearest 0.1 cm.

If the Bostwick viscosity value of the gravy from any can in the sample fails to conform to the requirement specified in 3.6, it shall be classified as a major defect and the lot shall be rejected.

4.5.4 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the U.S. Standards for Condition of Food Containers except that inspection for labeling shall be in accordance with 4.5.4.1. In addition, scratches, scuffs, or abrasions that occur on the outside coating as a result of the filling, sealing, and thermoprocessing of the tray pack cans shall not be scored as a defect.

4.5.4.1 Can label examination. Labels shall be examined for defects in accordance with MIL-L-1497 (see 5.4) except, for self-adhering labels, the following additional defects shall apply:

Major:	Label torn or scratched so as to obliterate any of the markings.
Minor:	Air bubbles under label. Label not properly adhered to can, for example, label raised or peeled back from edges or corners.

4.5.4.2 Label adhesive examination. When self-adhering labels are used, the adhesive shall be tested in accordance with ASTM D 3330.

4.5.5 Can closure examination. Can closures shall be examined visually and by teardowns in accordance with the can manufacturer's guidelines/ requirements and 21 CFR, Part 113, Subpart D or 9 CFR, Part 318, Subpart G, as applicable. Any nonconformance based on observation of can seam teardowns or on record of can seam teardowns shall be classified as a major defect and shall be cause for rejection of any involved product.

4.5.6 Vacuum examination. Cans shall be allowed to cool to 70°F ± 5°F, held for at least 24 hours after sealing, and then examined for vacuum retention. To examine, lay a straight edge in the center of the lid along

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the length of the tray pack. Both ends of the straight edge shall touch the lid at the inside edge of the double seam. There shall be a visible gap between the straight edge and the lid for the entire distance of the label panel. Using a shorter straight edge, the same procedure shall be used across the width, in the center of the tray pack can. One measurement shall be made when examining a ribbed lid, lay the straight edge only between the two center ribs along the length of the can. The inspection lot shall include only tray packs produced in a single shift on a single sealing machine. The sample size shall be 50 cans. Any nonconformance shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.7 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

Major:	National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible. Reinforced with other than nonmetallic strapping or tape. Dimensions of pads not as specified. Interior packing with fiberboard liner or pads not as specified.
Minor:	Other required markings missing, incorrect, or illegible. Arrangement or number of cans not as specified.

4.5.8 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PACKAGING

5.1 Preservation. The product shall be preserved in accordance with level A.

5.1.1 Level A. One hundred and six ounces of food product shall be filled into a tray pack can conforming to MIL-C-44340 and sealed and thermoprocessed as specified in 3.4 and 3.5.

5.2 Packing. The product shall be packed in accordance with level A, B, or C, as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L or HSC-L with an HSC full depth cover, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box, with the first two cans placed with the lids together and the next two

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cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads fabricated of grade V3c fiberboard. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). Flute direction of the box liner shall be vertical. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box. Each box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament-reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1), except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern. Boxes may be stacked by interlocking and reversing each tier, or by columnar stacking with paperboard or fiberboard sheets placed between each tier. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Four cans of product, preserved as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.3 Level C packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box, with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). Flute direction of the box liner shall be vertical. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of class domestic, grade 175 fiberboard.

5.3 Unit loading. When specified (see 6.1), the product, packed as specified in 5.2.2 or 5.2.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified, except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern. Boxes may be stacked by interlocking and reversing each tier, or by columnar stacking with paperboard or fiberboard sheets placed between each tier. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.4 Labeling. Each tray pack can shall be labeled in accordance with MIL-L-1497 and with the following:

- Official establishment number (for example, EST 38) or a three-letter code identifying the establishment

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- Lot number 1/

- Production shift number 1/

- Retort identification number 1/

- Retort cook number 1/

1/ The lot number shall be expressed as a four digit Julian code. The first digit shall indicate the year of production and the next three digits shall indicate the day of the year (for example, March 19, 1992 would be coded as 2078). The Julian code shall represent the day the product was packaged and processed. Sub-lotting (when used) shall be represented by an alpha character immediately following the four digit Julian code. Following the four digit Julian code and the alpha character (when used), the other required code information shall be printed in the sequence as listed above.

In addition, the name of the product shall be marked, stamping is permitted, on one 1001 by 200 end of the can. The labeling shall be legible when examined as specified in 4.5.4.1 after preparation of the product in accordance with heating instructions. Paper labels are not permitted. In addition, cans shall show the following statements:

TO HEAT IN WATER: Submerge unopened can in boiling water. Simmer gently 40 to 45 minutes. Avoid overheating (can shows evidence of bulging).

CAUTION: Use care when opening as pressure may have been generated within the can.

TO HEAT IN OVEN: Either punch several holes in the lid of the can or open the can in the usual manner leaving the loose lid in place. Place in a 350°F oven for 35 to 40 minutes.

WARNING: Do not place unopened can in oven. This may cause the can to burst.

YIELD: Serves 18 portions of 2/3 cup each.

As an alternate labeling method, a preprinted, self-adhering, 0.002 inch thick, clear polyester label printed with indelible black ink may be used. Self-adhering labels shall be applied after retorting. Pressure-sensitive adhesive shall require no preparation prior to application. Labels shall tack quickly and adhere without curling or breaking. The adhesive shall have a minimum adhesion of 60 ounces per inch width when examined as specified in 4.5.4.2. When self-adhering labels are used, the tray pack cans shall be labeled with the Julian code and a product code prior to retorting.

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5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, the following precautionary marking in capital letters larger than other markings shall be included:

CAUTION: DO NOT STACK PALLETS IN TRANSIT, OR MORE THAN TWO HIGH IN STORAGE, UNLESS PALLET RACKS ARE USED.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.4., and 6.2).
d. Provisions for approved preproduction samples (see 3.6.1 and 6.2).
e. Level of packing required (see 5.2).
f. Type and class of unit load when unit loading is required (see

5.2.1 and 5.3).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Appropriate level of pack. Based on conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.4 Cooking process. Cooking of the beef strips (see 3.3.1) to approximately 70 percent to 75 percent yield has been found adequate in complying with the finished product requirement (see 3.6).

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6.5 Beef patty preparation. It has been found that a patty mixture [illegible]

6.5 Alternate viscosity testing method. The contracting officer may authorize an alternate contractor-recommended method of viscosity testing if the alternate method is approved by the U.S. Army Natick Research, Development, and Engineering Center.

6.6 Subject term (key word) listing.

Canned food
Combat field feeding
Food processing
Operational ration
Shelf stable

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Review Activities:

Army - MD, QM
Navy - MC
DLA - SS

Preparing activity:

Army - GL

Project 8940-0751

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-B-44228B	1993 September 1

3. DOCUMENT TITLE

BEEF STRIPS WITH GREEN PEPPERS AND GRAVY, THERMOSTABILIZED, TRAY PACK

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON/DSN
U.S. Army Natick RD&E Center	508-651-4501	256-4501
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-WRE Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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